INCH-POUND

MIL-S-70682 (AR)
30 March 1990

MILITARY SPECIFICATION

SIMULATOR, FLASH, ARTILLERY , M21
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts for and loading, assembling and packing for one type of simulator designated as Simulator, Flash Artillery, M21.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	-Preservation - Packaging, Methods of
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370

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STANDARDS

MILITARY

MIL-STD-105	–Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	-Fuze and Fuze Components, Environmental and Performance Test For
MIL-STD-1234	-Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1235	-Single and Multi-Level Continuous Sampling Procedure

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9322059	-Simulator, Flash, Artillery, M21
9342970	-Packing and Marking for Box, Paperboard, for Simulator, Flash, Artillery, M21
9342972	-Packing and Marking for Box, Packing, for Simulator, Flash, Artillery, M21

(Copies of drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings, specifications and standards.

3.2 Assembly. The Simulator Assembly shall comply with all requirements specified on Drawing (dwg.) 9322059, and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 Charge composition. The moisture content of the charge composition at the loading station at the time of loading shall not exceed 0.10 percent.

3.3.2 Paper and paper products. The moisture content of the paper and paper products components of the simulator assembly at the loading station at the time of loading shall not exceed 6.0 percent.

3.4 Functioning (Ambient temperature, 21 ± 3°C, 70 ± 5°F, Hot 51.5 ± 3°C, 125 ± 5°F, Cold -32 ± 3°C, -25 ± 5°F). The assembly shall function and comply with the following requirements after conditioning at specified temperature.

3.4.1 Sound characteristics. The sound level intensity of the simulator shall be a minimum of 124 decibels (db) min. to 147 decibels (db) max. when measured at a distance of 96 ± .5 meters from the item under test.

3.4.2 Functioning time. The time from application of twenty-four (24) plus or minus (±) two (2) Volts Direct Current (VDC), with minimum ignition current of 3.0 milliwatt seconds/ohm, to initiation of the assembly shall not be greater than 0.50 seconds.

3.4.3 Flash characteristics. The flash intensity of the simulator shall be 4 x 10^6 candlepower minimum (min.). Residual smoke shall be white in appearance.

3.4.4 Heat soak at 160°F. (First Article Test Only) The packaged simulators (in barrier bag) shall meet the functioning requirements of 3.4.1, 3.4.2, and 3.4.3 after being subjected to a temperature of 71.1 ± 3°C, 160 ± 5°F for a minimum period of four (4) weeks. Prior to functioning, the simulator assembly resistance shall not exceed 2.0 ohms maximum.

3.4.5 Temperature - humidity. (First Article Test Only) The packaged simulators (in barrier bag) shall meet the functioning requirements of 3.4.1, 3.4.2, and 3.4.3 after exposure to temperature - humidity as specified in 4.5.3.5. Prior to functioning, the simulator assembly resistance shall not exceed 2.0 ohms maximum.

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3 .4.6 Functional timing sequencing of the electric igniter. (First Article Test Only) The continuity of the electric igniter shall be 10 milliseconds (ms) minimum at 12 VDC (nominal) at 0.5 amperes (amp). It shall produce an open circuit (if only for an instant) within 25 ms of receiving 12 VDC at 1 ampere when tested as specified in 4.5.3.6.

3.5 Jumble. The simulator shall comply with the following requirements.

a. The simulator shall not function during test.

b. There shall be no powder leakage.

c. The simulator components shall not separate during test.

3.6 Water immersion test. The simulator shall be capable of withstanding exposure to water immersion as specified in 4.5.6 and shall meet the functioning requirements of 3.4.1, 3.4.2, and 3.4.3.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

4. QUALTIY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Cap, Protecting	9342132	5/cavity
Container, Flash Charge	9342142	5/cavity
Body	9342143	5/cavity
Tube Assembly	9342144	10
Tube	9342145	25
Igniter, Electrical Assy (less quickmatch)	9342148	15
Cover, Flash Charge	9342149	5/cavity
Quickmatch	9342150	25
Cable, Electrical	9342153	10
Plug, Electrical Assembly	9342154	5/cavity
Insert, Threaded	9342155	25
Pin, Electrical	9342156	25
Cap, Short Circuit Assembly	9342157	5/cavity
Match, Electrical	9354514	15
Match and Cable Assembly	9354515	15
Packing and Marking	9342970	18
Packing and Marking for Box	9342972	1 box/162 items

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Simulator, Flash, Artillery M21 and Components		SHEET 1 OF 5		DRAWING NUMBER: See below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cap, Protecting (Dwg. 9342132)				
	Examination for defects	5/ cavity	-	3.2	4.4.2.1
	Container, Flash Charge (Dwg. 9342142)				
	Examination for defects	5/ cavity	-	3.2	4.4.2.2
	Tube (Dwg. 9342145)				
	Examination for defects	25	-	3.2	4.4.2.3
	Moisture content	(a)	-	3.3.2	4.5.2.2
	Tube Assembly (Dwg. 9342144)				
	Examination of defects	10	-	3.2	4.4.2.5
	Cable, Electrical (Dwg. 9342153)				
	Examination of defects	10	-	3.2	4.4.2.4

NOTES:

(a) Sample for this test shall be taken from test samples.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery M21 and Components		SHEET 2 OF 5		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9342143) Examination for defects	5/ cavity	-	3.2	4.4.2.6
	Cover, Flash Charge (Dwg. 9342149) Examination for defects	5/ cavity	-	3.2	4.4.2.8
	Quickmatch (Dwg. 9342150) Examination for defects	25	-	3.2	4.4.2.9
	Pin, Electrical (Dwg. 9342156) Examination for defects	25	-	3.2	4.4.2.10
	Insert, Threaded (Dwg. 9342155) Examination for defects	25	-	3.2	4.4.2.11

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Simulator, Flash Artillery M21 and Components		3 OF 5	See Below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Plug, Electrical Assembly (Dwg. 9342154)				
	Examination for defects	5/ cavity	-	3.2	4.4.2.12
	Cap, Short Circuit (Dwg. 9342157)				
	Examination for defects	5 cavity	-	3.2	4.4.2.14
	Igniter, Electrical Assy (Dwg. 9342148)				
	Examination for defects	15	-	3.2	4.4.2.15
	Electrical resistance test (less quickmatch)	15	-	3.2	4.4.2.15
	Functional timing sequence test less quickmatch)	15	-	3.4.6	4.5.3.6
	Match Electrical (Dwg. 9354514)				
	Examination for defects	15		3.2	4.4.2.16
	Packing and Marking For Box, Packing (Dwg. 9342972)				
	Examination for defects	1 box (162 items)	-	3.2	4.4.2.24

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER
	Simulator, Flash, Artillery M21 and components				See below
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Match and Cable Assembly (Dwg. 9354515)				
	Examination of defects	15		3.2	4.4.2.17
	Electrical resistance test (less quickmatch)	15	-	3.2	4.4.2.17
	Simulator, Flash, Artillery M21 (Dwg. 9322059)				
	Examination for defects	162	-	3.2	4.4.2.21
	Heat soak	108	-	3.4.4	4.5.3.4
	Water immersion	108	-	3.6	4.5.6, Note 1
	Functioning (Cold temperature)	36	-	3.4	4.5.3
	Sound characteristics		-	3.4.1	4.5.3.1
	Function Time		-	3.4.2	4.5.3.2
	Flash characteristics		-	3.4.3	4.5.3.3
	Functioning (Hot temperature)	36	-	3.4	4.5.3
	Sound characteristics		-	3.4.1	4.5.3.1
	Function Time		-	3.4.2	4.5.3.2
	Flash characteristics		-	3.4.3	4.5.3.3

NOTES:
(1) Function testing to be performed after water immersion and temperatute conditioning.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-70682 (AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery M21 and Components		SHEET 5 OF 5		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Functioning (Ambient Temperature)	36	-	3.4	4.5.3
	Sound characteristics		-	3.4.1	4.5.3.1
	Function time		-	3.4.2	4.5.3.2
	Flash characteristics		-	3.4.3	4.5.3.3
	Temperature - humidity	18	-	3.4.5	4.5.3.5
	Resistance		-	3.4.5	4.5.3.5
	Sound characteristics		-	3.4.1	4.5.3.1
	Function time		-	3.4.2	4.5.3.2
	Flash characteristics		-	3.4.3	4.5.3.3
	Heat soak at 160°F	18	-	3.4.4	4.5.3.4
	Sound characteristics		-	3.4.1	4.5.3.1
	Function time		-	3.4.2	4.5.3.2
	Flash characteristics		-	3.4.3	4.5.3.3
	Jumble	18	-	3.5	4.5.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Simulator Assemblies shall contain:

a. Parts of one kind from one manufacturer.

b. Electric igniters from not more than one lot interfix number from one manufacturer.

c. Composition ingredients from not more than one lot or batch.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cap, Protecting				9342132
					NEXT HIGHER ASSEMBLY 9322059
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Large inside diameter (I.D.), minimum (min)		0.40%	3.2	Gage
102	Length of large I.D., min.		0.40%	3.2	Gage
Minor					
201	Total length		0.65%	3.2	Gage
202	Poor workmanship		1.00%	3.8	Visual

NOTES: *See 4.5.1 For Qualification and Production Control of Mold Cavities.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Container, Flash Charge				9342142
					NEXT HIGHER ASSEMBLY 9342141
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Outside diamater, maximum (max)		0.40%	3.2	Gage
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Poor workmanship		1.00%	3.8	Visual

NOTE: *See 4.5.1 For Qualification and Production Control of Mold Cavities.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9342145
4.4.2.3	Tube				NEXT HIGHER ASSEMBLY 9342144
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter, (min.)		0.40%	3.2	Gage
102	Outside diameter, (max.)		0.40%	3.2	Gage
103	Moisture content	Note 1		3.3.2	4.5.2.2
Minor					
201	Total length		0.65%	3.2	Gage
202	Poor workmanship		1.00%	3.8	Visual

NOTES:

(1) Sampling in accordance with 4.4.3.1.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.4	Cable, Electrical		1	1	9342153
					NEXT HIGHER ASSEMBLY 9354515

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length of short stripped end		0.65%	3.2	Gage
202	Length of long stripped end		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Poor workmanship		1.0%	3.8	Visual

NOTES:

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MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	Tube Assembly	1	1	9342144
				NEXT HIGHER ASSEMBLY: 9342141

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Plug above flush		0.40%	3.2	Gage
Minor					
201	Plug greater than specified dimension below flush		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.8	Visual

NOTES:

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MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Body		9342143
			NEXT HIGHER ASSEMBLY 9342141

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined	*			
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Corner to corner of flange		0.40%	3.2	Gage
103	Flange rib(s) missing or damaged		0.40%	3.2	Visual
Minor					
201	Small inside diameter		0.65%	3.2	Gage
202	Large inside diameter		0.65%	3.2	Gage
203	Total length, min.		0.65%	3.2	Gage
204	Draft angle missing or incorrect		0.65%	3.2	Gage
205	Poor workmanship		1.0%	3.8	Visual

NOTES:

*See 4.5.1 For Qualification and Production Control of Mold Cavities.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Case, Cartridge Assembly	1 OF 1	9342141
			NEXT HIGHER ASSEMBLY: 9342134

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	None defined				
Minor					
201	Any component(s) missing, damaged, or improperly assembled		0.65%	3.2	Visual/Manual
202	Poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.8	Cover, Flash Charge		1 OF 1	9342149
				NEXT HIGHER ASSEMBLY: 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Outside diameter of ribs, max		0.40%	3.2	Gage
102	Holes for cables missing or incorrect		0.40%	3.2	Gage
	Hole for quickmatch missing or incorrect		0.40%	3.2	Gage
Minor					
201	Total length (overall)		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.8	Visual
203	Ribs on cover missing or damaged		0.65%	3.2	Visual
204	Upper flange thickness incorrect		0.65%	3.2	Gage

NOTES:

*See 4.5.1 For Qualification and Production Control of Mold Cavities.

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MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Quickmatch		9342150
			NEXT HIGHER ASSEMBLY 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length		0.40%	3.2	Gage
102	Incorrect number of strands		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.8	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Pin, Electrical	1 OF 1	9342156
			NEXT HIGHER ASSEMBLY: 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of slotted end to flange		0.40%	3.2	Gage
102	Length to thread		0.40%	3.2	Gage
103	O.D. of slotted end		0.40%	3.2	Gage
104	Pitch diameter, min.		0.40%	3.2	Gage
105	Major diameter, min.		0.40%	3.2	Gage
Minor					
201	Total length		0.65%	3.2	Gage
202	O.D. of flange		0.65%	3.2	Gage
203	Presence of slot		0.65%	3.2	Gage
204	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Insert, Threaded	1 OF 1	9342155
			NEXT HIGHER ASSEMBLY: 9342154

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Knurl missing, incorrect or damaged		0.40%	3.2	Visual
102	Pitch diameter, max		0.40%	3.2	Gage
103	Minor diameter, max		0.40%	3.2	Gage
104	Protective finish missing		0.40%	3.2	Visual
Minor					
201	Total length		0.65%	3.2	Gage
202	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Plug, Electrical Assembly				9342154
					NEXT HIGHER ASSEMBLY 9342148
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Thickness of body, max		0.40%	3.2	Gage
102	Width under head (Datum B)		0.40%	3.2	Gage
103	Length under head (Datum A)		0.40%	3.2	Gage
104	Threaded insert missing or damaged		0.40%	3.2	Visual
105	True position of threaded insert (2)		0.40%	3.2	Gage
Minor					
201	Presence of molding compound, dirt, or foreign matter on or around insert threads		0.65%	3.2	Visual
202	Poor workmanship		1.00%	3.8	Visual

NOTES:

*See 4.5.1 For Qualification and Production Control of Mold Cavities.

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QUALTIY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Clip	1 OF 1	9342158
			NEXT HIGHER ASSEMBLY 9342157

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length of clip, min.		0.40%	3.2	Gage
102	Height of clip		0.40%	3.2	Gage
103	Thickness of clip		0.40%	3.2	Gage
104	Protective finish missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.00%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Cap, Short Circuit	1 OF 1	9342157
			NEXT HIGHER ASSEMBLY: 9342148

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	Inside dimension of cap opening, min		0.40%	3.2	Gage
102	Depth of cap opening		0.40%	3.2	Gage
103	True position of clip		0.40%	3.2	Gage
104	Inside radius of cap opening		0.40%	3.2	Gage
Minor					
201	Width O.D. of cap		0.65%	3.2	Gage
202	Total length of cap		0.65%	3.2	Gage
203	Poor workmanship		1.00%	3.8	Visual

NOTES:

*See 4.5.1 For Qualification and Production Control of Mold Cavities.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.15	Igniter, Electrical Assembly		1 OF 1	9342148
				NEXT HIGHER ASSEMBLY: 9342134

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total lenght, min.		0.40%	3.2	Gage
102	Electrical resistance test 1/		0.40%	3.2	Ohmmeter
103	Quickmatch insecure or improperly assembled		0.40%	3.2	Visual/Manual
104	Functional timing sequence test 1/	30	0-1	3.4.6	4.5.3.6
Minor					
201	Total length, max		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.8	Visual

NOTES:

1/ Less quickmatch

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Match Electrical	1 OF 1	9354514
			NEXT HIGHER ASSEMBLY: 9354515

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Total length		0.65%	3.2	Gage
202	Width at end of large angle		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Match and Cable Assembly				9354515
					NEXT HIGHER ASSEMBLY 9342148
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length to electrical match		0.40%	3.2	Gage
102	Electrical resistance test		0.40%	3.2	Ohmmeter
103	Match insecure or improperly assembled (soldered)		0.40%	3.2	Visual/Manual
Minor					
201	Poor workmanship		1.00%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Simulator, Flash, Artillery M21, Loading Assembly (Prior to assy of Detonation Charge Cover)	1 OF 1	9342134
			NEXT HIGHER ASSEMBLY 9322059

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of Charge (Verification)	50		3.2	Balance(Note a or b)
102	Composition-Moisture Content	Note 1		3.3.1	4.5.2.1
Minor					
201	Poor workmanship		1.0%	3.10	Visual

(a) A sample of fifty (50) loading assemblies, prior to sealing the detonation charge cover shall be selected from each eight (8) hours of production and subjected to a weight of charge verification as follows:

Using a tared cartridge case (9342141), the weight difference due to the addition of the detonation charge is determined. If the weight of any charge fails to meet the applicable requirements, those assemblies loaded since the last acceptable verification test shall be classed defective and removed from the lot. (Non-Destructive Test)

(b) MIL-STD-1235 single and multi-level continuous sampling procedure may be used.

NOTES:

(1) Sampling in accordance with 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.19	Simulator, Flash, Arty, M21 Loading Assembly	1	1	9342134
				NEXT HIGHER ASSEMBLY 9322059

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Evidence of loose powder on exterior of assembly		100%	3.2	Visual
Major					
101	Assembly damaged to extent that powder may leak		0.40%	3.2	Visual
102	Igniter lead(s) pinched or improperly located		0.40%	3.2	Visual
103	Any component loose at any joint or contact		0.40%	3.2	Visual/Manual
Minor					
201	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Simulator, Flash, Artillery M21 (Prior to assembly of protecting cap)	1 OF 1	9322059
			NEXT HIGHER ASSEMBLY ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Short circuit cap missing or damaged		100%	3.2	Visual
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Simulator, Flash, Artillery M21	1 OF 2	9322059
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Leakage of explosive from assembly, or explosive (powder) on exterior of assembly		100%	3.2	Visual
2	Jumble	12	0-1	3.5	4.5.4
Major					
101	Sealing tape missing, peeling, or incorrectly applied		0.40%	3.2	Visual
102	O.D. of body (max.)		0.40%	3.2	Gage
103	Assembly damaged, dented or deformed		0.40%	3.2	Visual
104	Water immersion	50	1/	3.6	4.5.6
105	Functioning (Hot)	50	2/ 2-3		
	Sound characteristics		Ac Re	3.4.1	4.5.3.1
	Functioning time			3.4.2	4.5.3.2
	Flash characteristics			3.4.3	4.5.3.3
106	Functioning (Cold)	50	2/ 3-4		
	Sound characteristics			3.4.1	4.5.3.1
	Functioning time			3.4.2	4.5.3.2
	Flash characteristics			3.4.3	4.5.3.3

NOTES: 1/ Water immersion samples to be used in ambient function test (Major 107).
2/ After three (3) consecutive lots have been accepted, sample size may be reduced in accordance with 4.4.5.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9322059
4.4.2.21 (Cont)	Simulator, Flash, Artillery M21				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107	Functioning (Ambient)	50	2/ 2-3		
	Sound charcteristics			3.4.1	4.5.3.1
	Functioning time			3.4.2	4.5.3.2
	Flash characteristics			3.4.3	4.5.3.3
Minor					
201	Total length (max.)		0.65%	3.2	Gage
202	Marking missing, incorrect or unidentifiable		0.65%	3.2	Visual
203	Poor workmanship		1.0%	3.8	Visual

NOTES:
2/ After three (3) consecutive lots have been accepted, sample size may be reduced in accordance with 4.4.5.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Box, Paper Board, for Simulator, Flash Artillery, M21 (Prior to sealing top flaps)				9342970
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Component(s) missing		0.40%	3.2	Visual
102	Filler(s) missing or inadequate		0.40%	3.2	Visual
103	Packaging components obviously damp or wet		0.40%	3.2	Visual/Manual
Minor					
201	Label missing		0.65%	3.2	Visual
202	Marking incorrect, misleading or unidentifiable		0.65%	3.2	Visual
203	Instruction sheet missing		0.65%	3.2	Visual
204	Poor workmanship		1.00%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-S-70682(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Sealed Packed Barrier Bag, Prior to Packing Into Box	1 OF 1	9342970
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of bag destroyed by rupture, puncture, or separation of seal	10	0.40%	3.2	Visual
102	Heat seal of barrier bag incorrect		0-1	3.2	4.5.5
Minor					
201	Heat seal of bag located so that two subsequent seals cannot be made		0.65%	3.2	Visual
202	Marking on bag missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Poor workmanship		1.00%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-70682(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Box, Packing, For Simulator, Flash, Artillery, M21	1 OF 1	9342972
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
102	Box broken or split		0.40%	3.2	Visual
103	DOT symbol misleading or unidentifiable		0.40%	3.2	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	3.2	Visual
202	Hardware improperly engaged		0.65%	3.2	Visual
203	Strapping improperly assembled		0.65%	3.2	Visual
204	Metallic Seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
205	Contents loose		0.65%	3.2	Manual
206	Poor workmanship		1.00%	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article table and Quality Conformance Inspection.

4.4.3.1 Moisture content (see Table II) - major defects

TABLE II

Charge composition	(see 3.3.1)
Paper and paper products	(see 3.3.2)

The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall selects and subject to test, one sample of each material from each eight (8) hours production. If a composite sample exceeds the requirement, that quantity or sub-lot material represented by the sample shall not be used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.5 Functioning test.

4.4.5.1 Hot and ambient temperature. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.4.2.21, fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of 4.4.2.21, thirty-two(32) simulators shall be selected from each lot for this test. The lot shall be rejected in one (1) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.21.

4.4.5.2 Cold temperature. beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria of 4.4.2.21 fifty (50) simulator assemblies shall be selected from each lot for test. After three (3) consecutive lots have met the acceptance criteria of 4.4.2.21, thirty-two (32) simulators shall be selected from each lot for this test. The lot shall be rejected if two (2) or more major functioning defects occur, and lot sampling shall revert to fifty (50) samples per lot in accordance with 4.4.2.21.

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4.5 Methods of inspection.

4.5.1 Dimensional control of molded or plastic parts. In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1), a sample if three(3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. Each molded part shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.1, 4.4.2.2, 4.4.2.6, 4.4.2.8, 4.4.2.12 and 4.4.2.14 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective part(s) are found during qualification of the cavity; the cavity producing the defective part(s) will not be used in production. If any defective part(s) are found when inspection is performed for the control of the cavity, the cavity producing the defective part(s) shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be submitted per 6.3 for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time dimensionally check ten(10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold unitl dimensional stability is attained. The inspection data used in determining the curing time shall be sent to the procuring activity (reference 6.5). If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

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4.5.2 Moisture content.

4.5.2.1 Charge composition. The method of determing the percentage of moisture in the flash charge shall be in accordance with Method 102.1 of MTL-STD-1234 except that the sample shall be heated until a constant weight is obtained in lieu of four hours. Observation shall be made for compliance with the applicable requirement.

4.5.2.2 Paper and paper products. The moisture content shall be determined in accordance with Method 102.1.1 of Standard MIL-STD-1234 except that a ten(l0) gram sample shall be used and the drying time shall be for two(2) hours.

4.5.3 Functioning. (Ambient temperature, 21 + 3°C, 70 ± 5°, Hot, 51.5 + 3°C, 125 + 5°F, Cold, -32 + 3°C, -25 + 5°F). The simulator assemblies shall be conditioned for 12 hours min. at the specified temperatures and fired within 15 minutes after removal from the conditioning chamber. The simulator shall be mounted horizontally in an approved fixture (simulating the Hoffman tube tank gun fire simulator with 9 firing drums) on a rigid support and fired by means of an approved power source in accordance with 3.4.2. (Destructive test).

4.5.3.1 Sound characteristics. Measurement shall be made from a side position to the longitudinal axis of the simulator which is to be suspended horizontally 4.9 ± .20 meters off the ground. The detector shall be positioned 1.5 ± 0.1 meters off the ground for measurement purposes. Observation shall be made for compliance with the applicable requirement.

4.5.3.2 Functioning time. The time from initiation of the VDC shall be measured with approved metering equipment in accordance with para. 4.4.4. The assembly shall be observed for compliance with the applicable requirement.

4.5.3.3 Flash characteristics. The flash intensity of the simulator shall be recorded with approved method in accordance with para. 4.4.4. The simulator and detector positioning shall be the same as para. 4.5.3.1. Observation will be made for conformance with the applicable requirements.

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4.5.3.4 Heat soak at 160°F. (First Article Only) The packaged simulator assemblies shall be conditioned for a period of four(4) weeks min. at the specified temperature. After removal from the conditioning chamber and allowed to return to ambient temperature, resistance measurement shall be made using a Fluke 8024B, 8050A Multimeter, or equivalent meter, approved for use with electro-explosive devices.

SAFETY NOTE: All resistance measurements shall be conducted from behind a barricade, and shorting caps shall be in place at all times except during resistance measurement and when functioning items.

The simulators resistance shall not exceed the applicable requirement after conditioning. Only those simulators passing the resistance requirement shall be placed in the firing device and functioned (ref. 4.5.3.1, 4.5.3.2, 4.5.3.3). Any simulator failing to meet the applicable requirements shall be classed defective. (Destructive test)

4.5.3.5 Temperature - humidity. (First Article Test Only) The packaged simulators shall be subjected to a 28 day temperature - humidity test in accordance with MIL-STD-331, Test 105.1, para. 5.2, Single Chamber Cycling Method. After removal from the conditioning chamber and allowed to return to ambient temperature, resistance measurements shall be made and the simulators functioned as specified in 4.5.3.4 above. Any simulator failing to meet the applicable requirements shall be classed defective. (Destructive test).

4.5.3.6 Functional timing sequencing test. (First Article Test Only) A continuity and open circuit test shall be performed on the electric igniter using an approved method and equipment in accordance with 4.4.4. observation shall be made for compliance with applicable requirement. (Destructive test)

4.5.4 Jumble. The simulator assemblies shall be tested in accordance with procedures specified in Standard MIL-STD-331, Test 102, except that the duration of the test shall be a minimum of 20 minutes. The assemblies and equipment shall be observed for compliance with the applicable requirements. (Destructive Test)

4.5.5 Heat seal of barrier bag. The test shall be performed in accordance with the procedure specified in MIL-P-116H para. 4.4.4.3. After the test, the seal shall be examined for compliance with the applicable requirements. (Non-Destructive Test).

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4.5.6 Water immersion test. The unpackaged simulator, shall be submerged in 12 inches minimum of water at ambient (70 + 5°F) temperature for a minimum of one hour, The simulator shall then be removed from the water, wiped off, temperature conditioned in accordance with 4.5.3, and placed in the firing device, and functioned per 4.5.3.1, 4.5.3.2 and 4.5.3.3.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on Dwg . 9342970.

5.2 Packing and marking. Packing and marking shall be as specified on Dwgs. 9342970 and 9342972.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The components covered by this specification are intended for use on the Simulator, Flash, Artillery, M21.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Distribution of ammunition data cards. Distribution of data cards include the following: Commander, US Army Armament Research and Development and Engineering Center, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, NJ 07806-5000.

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6.6 Subject term (key word) listing.
Candle power
Charge composition
Electric Igniter
Flash Simulator

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1370-A291)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-70682 (AR)	30 MARCH 1990

3. DOCUMENT TITLE
SIMULATOR, FLASH, ARTILLERY, M21 PARTS FOR AND LOADING, ASSEMBLING & PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY, ARDEC STANDARDIZATION & SPECIFICATION OFFICE	(1) Commercial (201)724-6671	(2) AUTOVON 880-6671
ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S, BLDG 6 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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